

1. Research Questions

- Q1 (One-Way ANOVA): Is there a significant difference in the average daily total screen time among students with different physical fighting frequencies?
- Q2 (Two-Way Extension): Does the impact of physical fighting frequency on total screen time significantly depend on the student's gender?

2. Data & Variables

Dataset: YRBS 2007 (N = 13,155 after listwise deletion)

- Grouping Variable 1 (X1): **Fighting_Group** (1_None, 2_Occasional, 3_Frequent)
- Grouping Variable 2 (X2): **WhatIsYourSex** (Female, Male)
- Response Variable (Y): **Total_Screen_Time** (Continuous; combined daily TV and computer hours)

4. Method Selection

1. One-Way ANOVA
 - Purpose: Tests if the mean daily screen time significantly differs across the three fighting groups. Post-hoc Test: Tukey's HSD to identify specific significant pairs.
2. Two-Way ANOVA with Interaction
 - Purpose: Investigates if the main effects of sex and fighting group, along with their simultaneous interaction effect, hold statistical significance. Model Formula: Total_Screen_Time fits Fighting_Group multiplied by Sex.

3. Exploratory Data Analysis (EDA)

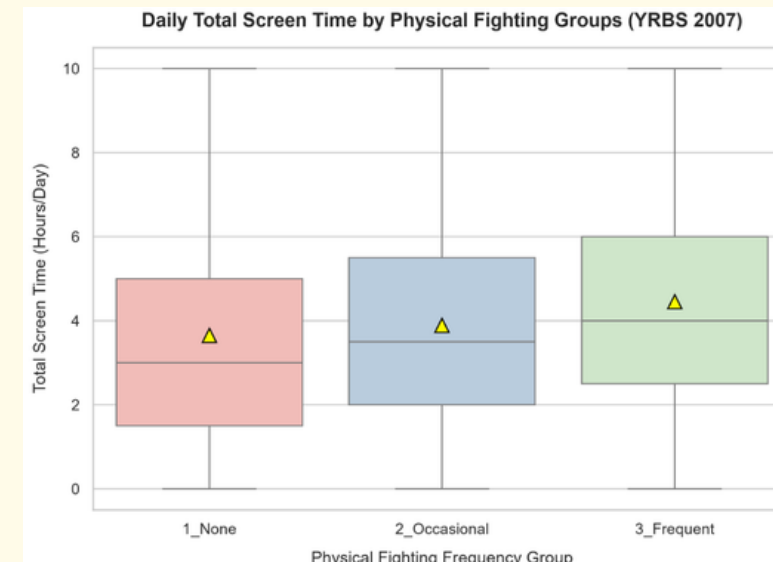


Figure 1:
Boxplot of Screen Time by Fighting Group

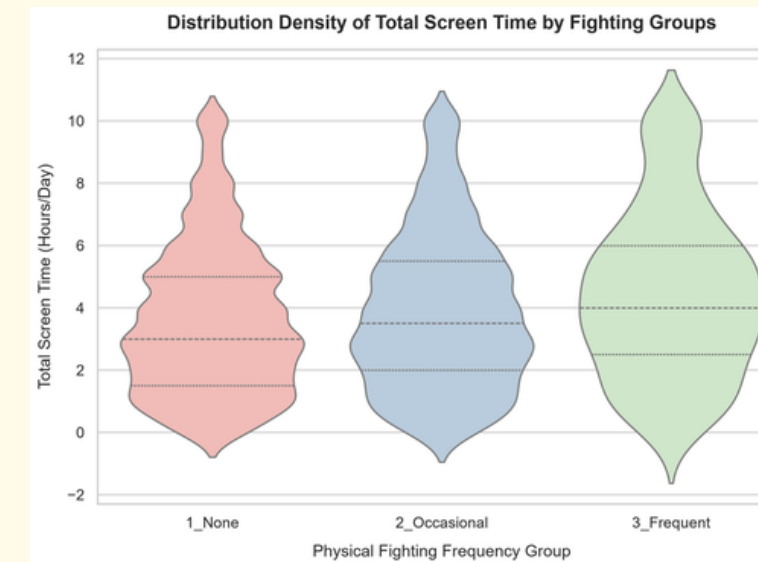


Figure 2:
Violin Plot of Screen Time Distribution

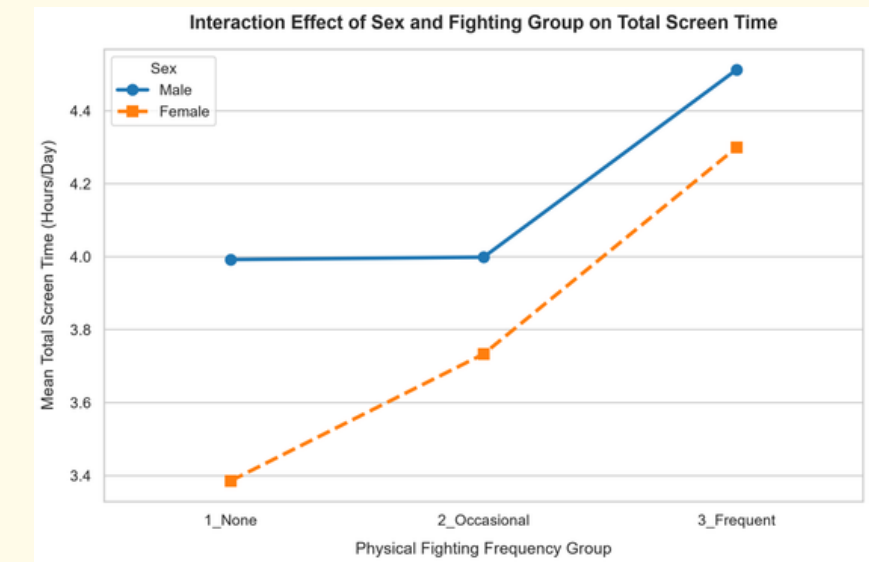


Figure 3:
Interaction Plot of Sex and Fighting Group

5. Key Results: One-Way ANOVA

Sample Estimates

- 1_None Mean = 3.65 hours/day (n = 8,628)
- 2_Occasional Mean = 3.89 hours/day (n = 4,133)
- 3_Frequent Mean = 4.45 hours/day (n = 394)

Test Inference

- F-statistic = 21.03 (Highly significant)
- **P-value < 0.001**
- Decision: **Reject H_0** at $\alpha = 0.05$
- Tukey's HSD: All pairwise group differences are statistically significant.

6. Key Results: Two-Way & Interaction (Extension)

Subgroup Trajectories (Female vs. Male)

- Female Mean: 1_None (3.39h) -> 2_Occasional (3.73h) -> 3_Frequent (4.30h)
- Male Mean: 1_None (3.99h) -> 2_Occasional (4.00h) -> 3_Frequent (4.51h)

Two-Way ANOVA Effects

- Main Effect of Sex: $p < 0.001$ (Significant)
- Main Effect of Fighting: $p < 0.001$ (Significant)
- Interaction Effect (Sex x Fighting): **$p < 0.05$ (Statistically Significant)**

7. Final Conclusion

Statistical Decisions

- One-Way ANOVA: **Reject H_0** ($\alpha = 0.05$). Fighting frequency has a significant impact on screen time.
- Two-Way Interaction: **Reject H_0** ($\alpha = 0.05$). The effect of fighting on screen time depends on gender.

The Core Meaning

- "Adolescent conflict scales positively with digital screen exposure. However, this behavioral pathway is gender-heterogeneous. While females display a steady, linear increase in screen hours with any level of conflict involvement, males experience a stagnant baseline between non-fighters and occasional fighters, followed by an intense spike in screen time among the frequent fighting cohort."